

pyunicorn v1.0: A versatile toolbox for complex system analysis across disciplines

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Summary

The pyunicorn (Unified Complex Network and Recurrence Analysis) toolbox provides a unique collection of data-driven assessment methods for complex system phenomena — including various aspects of dynamical complexity, spatio-temporal interrelations, extreme events and critical transitions. Its algorithms combine network theory and nonlinear time series analysis, with a focus on constructing and quantifying the following types of numerical objects:

- Time series networks** (e.g., recurrence networks, visibility graphs): Within a time series, a pairwise relation is computed over states/events/patterns, which are in turn defined by a predicate over time intervals.
- Functional networks** (e.g., climate networks): Within a collection of mutually dependent time series, members are linked according to some measure of statistical association.
- Surrogate networks/time series** (e.g., twin surrogates): Samples are drawn from a conditionally uniform distribution over networks/time series, under some structural constraints motivated by hypothesis testing purposes.

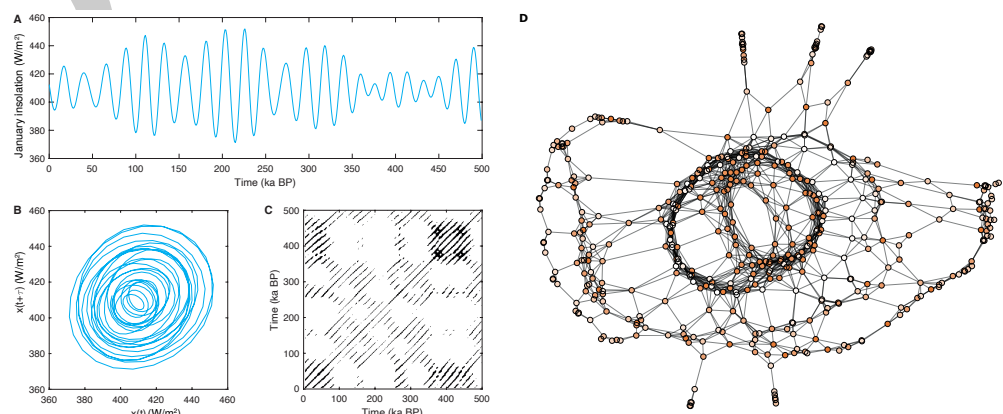


Figure 1: Example of a recurrence network in climatology (Marwan et al., 2021)¹. (A) Time series of January insolation at latitude 20 °N. (B) Delay coordinate embedding for A, with embedding dimension $m = 2$ and delay $\tau = 6$ ka. (C) Recurrence plot for B, with recurrence threshold $\epsilon = 10$ W/m². (D) Recurrence network for C, with nodes of darker colour representing later points in time.

pyunicorn is implemented in Python and C/Cython (Behnel et al., 2011), and was first published over a decade ago (Donges, Heitzig, et al., 2015). Since then, it has been maintained as an open source community project, benefitting from numerous user requests, new features, tutorials, corrections and refactorings. A growing interdisciplinary user base, as well as fluctuating levels of funding and development over the years, have warranted a recent concerted effort to consolidate the package. The present paper accompanies the resulting first major version release², and compiles applications published to date.

Statement of need

Network theory and dynamical system theory have long provided two complementary perspectives on complex systems: The former analyses the structure of interactions (links/edges) between subsystems (nodes/vertices), whereas the latter characterises systemic behaviour in time, such as predictability and chaos, bifurcations and regime shifts. When the development of pyunicorn began in 2008, these two fields had started growing together specifically along two strands of research.

On the one hand, once a time series observing a complex system has been represented as a **time series network**, dynamical properties of the system can be investigated using network theory methods. Various linking criteria have been proposed for the construction of time series networks, including recurrence relations within trajectories in phase space or embedding space (Donges et al., 2012; Marwan et al., 2007, 2009) (cf. Figure 1), and visibility relations within function graphs of scalar time series (Lacasa et al., 2008).

On the other hand, linear or nonlinear statistical association measures from the time series analysis literature can be used to construct a **functional network** from a collection of time series, which yields a topological description of functional interdependence. This methodology was especially put forward in neuroscience (Bullmore & Sporns, 2009; Zhou et al., 2006, 2007) and climatology (Donges et al., 2009; Donges, Petrova, et al., 2015), and has found further applications in fields such as economics and finance (Huang et al., 2009).

In addition to accompanying these syntheses, the pyunicorn library has also facilitated the development of a number of more specialised methods by affiliated research groups; and for many such methods, it remains the only actively maintained open source implementation to date. The emergence of many new interdisciplinary applications (cf. Section 5) has further affirmed the claim that, beyond its origin in climate and Earth system science, pyunicorn's network approach is "widely applicable in numerous fields" (Donges, Heitzig, et al., 2015).

State of the field

Long-established packages in the Python ecosystem, such as `networkx` (Hagberg et al., 2008), `python-igraph` (Csárdi & Nepusz, 2006) and `networkit` (Angriman et al., 2022), extensively cover graph algorithms. In addition, `graph-tool` (Peixoto, 2014, 2023) implements nonparametric Bayesian methods for hierarchical community detection. Libraries with a stronger methodological focus include: `PyRQA` (Rawald et al., 2017) and `AccRQA` (Adámek et al., 2026) for recurrence quantification analysis on large datasets, `ordpy` (Pessa & Ribeiro, 2021) for time series analysis with ordinal networks, `smt` (Saves et al., 2024) and `irreversibility` (Zanin, 2025) for surrogate modelling and for irreversibility tests of time series, and `dynamicalsys` (Sales et al., 2025) for dynamical system analysis. Siebert et al. (2021) provide a systematic review of Python packages for various time series analysis applications. Of course, comparable software also exists for other programming languages, including: `CRP Toolbox` (Marwan, 2024) and `NoLiTiA` (Weber & Oehrn, 2022) in Matlab, `DynamicalSystems.jl` (Datseris, 2018) in Julia, and `nonlinearTseries` (Garcia & Sawitzki, 2026) in R.

¹Reprinted from Marwan et al. (2021) with permission from Elsevier. Original under CC BY-NC-ND 4.0.

²Cf. Semantic Versioning specification.

Yet, pyunicorn has maintained its unique position, building on established graph packages to bridge and complement the above application areas. For instance, pyunicorn.Network is implemented by extending python-igraph.Graph with more advanced constructs, such as *coupled* or *multilayer* networks (Donges et al., 2011) and *node-weighted* or *node-splitting-invariant* network measures (Heitzig et al., 2012; Wiedermann et al., 2013; Zemp et al., 2014). Overall, pyunicorn's strength lies in its integration of methods from complex network theory and nonlinear time series analysis. Zou et al. (2019), Silva et al. (2021) and Marwan et al. (2021) provide thorough summaries of relevant concepts and algorithms.

Software design

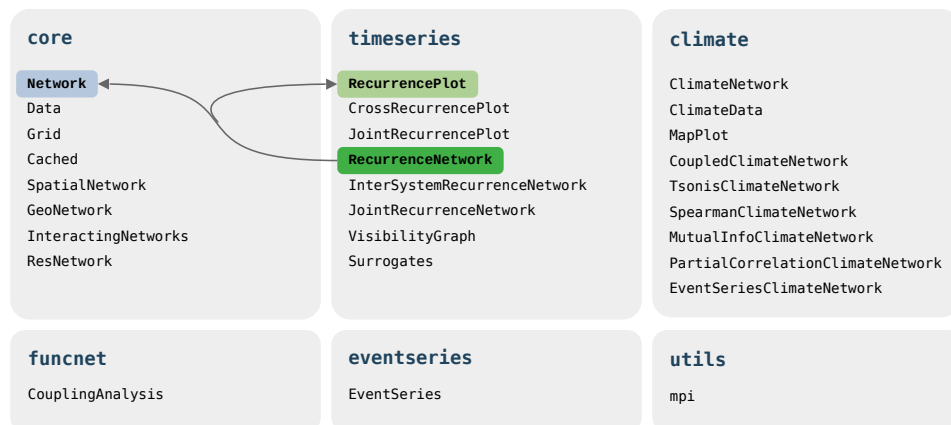


Figure 2: Overview of pyunicorn's modules/classes, with an example class inheritance relation.

pyunicorn follows an object-oriented design, with module and class hierarchies that isolate core data structures and reflect relationships between analysis methods. For example, RecurrenceNetwork inherits much of its functionality from RecurrencePlot and Network (cf. Figure 2).

pyunicorn.Network is the central class, and it is backed both by a python-igraph graph (Csárdi & Nepusz, 2006) and by a scipy.sparse matrix (Virtanen et al., 2020). In addition to importing, constructing, querying, updating and exporting these internal data structures, Network also implements more specialised network annotations, network measures, and random graph models. Most other classes in the package derive from Network, extending it with dedicated logic for analyses. Many library methods are implemented by calling into pyunicorn's own C/Cython extension modules via array interfaces.

Overall, this design covers many common use cases in complex system science: It supports data imports from various formats, its analysis methods are easy to navigate, and it integrates naturally with the Python numerical computing ecosystem. While out-of-core and high-performance computing were not primary design goals, pyunicorn's graph algorithms achieve a performance level that is comparable to, or competitive with, many other general-purpose implementations.

The library has tracked changing standards in package management, extension compilation, linting, testing and continuous integration, and it is compatible with all officially supported Python versions on Windows, macOS and Linux. For improved maintainability, the Python/Cython interface was overhauled, and most extension modules were ported from C to Cython. Substantial additions to the test suite, as well as removal of experimental and untested code, have increased test coverage from less than 50% to nearly 80%.

pyunicorn has also grown by a range of new or improved functionality (cf. [Changelog](#)):

- Many measures in Network, and especially their node-splitting-invariant refinements, were generalised to support weighted and directed networks.
- Spatial and coupled network analyses were generalised, by adding SpatialNetwork, by adding a Watts-Strogatz model to Network, and by adding new measures to RecurrencePlot, CoupledClimateNetwork and InteractingNetwork.
- EventSeries was added along with EventSeriesClimateNetwork, and subsequently received several algorithmic improvements.
- MapPlot was refactored and simplified.
- State management and memoisation were consolidated via the new mix-in Cached.
- A collection of tutorial notebooks was added to the documentation.

Research impact statement

Since its initial release, pyunicorn has found numerous applications across the sciences³:

- Astrophysics, e.g., active galactic nuclei ([Phillipson et al., 2023](#)), black holes ([Broadbent & Phillipson, 2024](#)) and stars ([George et al., 2020](#)).
- Geophysics, e.g., crustal displacements ([Hobbs & Ord, 2018](#)), seismic tomography ([Talavera-Soza & Deuss, 2023](#)) and magnetospheric dynamics ([Donner et al., 2019](#)).
- Climate and paleoclimate science ([Caesar et al., 2020](#); [Donges, Donner, et al., 2015](#); [Franke et al., 2017](#); [Franke & Donner, 2017](#); [Haas et al., 2023](#); [Jiang et al., 2024](#); [Lekscha & Donner, 2020](#); [Li et al., 2026](#); [Marwan et al., 2021](#); [Moinat et al., 2024](#); [Nocke et al., 2015](#); [Sumit et al., 2026](#); [Sun et al., 2018](#); [Wiedermann et al., 2017](#); [Wolf, Ozturk, et al., 2021](#); [Wolf, Voigt, et al., 2021](#); [Wolf & Donner, 2021](#)), and in particular El Niño related phenomena ([Broni-Bedaiko et al., 2019](#); [Ekhtiari et al., 2021](#); [Feng & Dijkstra, 2017](#); [Oluwole, 2017](#); [Wiedermann, Radebach, et al., 2016](#)).
- Social science, e.g., societal clustering and marginalisation ([Schleussner et al., 2016](#)).
- Biological and medical research, e.g., mammal evolution ([Bekeraité et al., 2025](#)), cancer research ([Amemiya et al., 2025](#)), behavioural neuroscience ([Maksimenko et al., 2019](#)) and EEG data ([Frolov et al., 2023](#); [Lechner & Northoff, 2024](#)).
- Engineering, e.g., warning signals for thermoacoustic instabilities ([Waxenegger-Wilfing et al., 2021](#)) and functional dependence in coupled oscillators ([Geier et al., 2024](#)).
- Particle physics, e.g., collision event analysis ([Mullin et al., 2021](#)).
- Methodological advances in network theory and nonlinear dynamics ([Alberti et al., 2020](#); [Lekscha & Donner, 2018](#); [Medrano et al., 2021](#); [Odenweller & Donner, 2020](#); [Sales et al., 2023](#); [Silini & Masoller, 2021](#); [Subramaniam et al., 2015](#); [Wiedermann, Donges, et al., 2016](#); [Yuan & Shou, 2024](#)).

Furthermore, pyunicorn's tutorial notebooks provide introductions to various complex system analysis methodologies, and have been repeatedly employed in academic teaching.

AI usage disclosure

No generative AI was used in the development and maintenance of this software, or in the writing of this manuscript. The use of AI coding assistance is not prohibited for contributors.

Author contributions

Software: FK coordinated maintenance and releases in recent years. BB contributed to design and maintenance, provided technical guidance, and edited tutorials. MB contributed to maintenance and wrote tutorials. JFD and RVD organised funding and supervised development.

³26 out of the 46 surveyed publications are unaffiliated with the package authors.

149 *Manuscript*: FK and BB developed the full draft. MB drafted [Section 5](#). MB, JFD and RVD
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